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Universal collapse and sex-dependent scaling laws in human performance and aging

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ABSTRACT

Human performance declines with age, yet whether this process follows universal scaling laws across heterogeneous populations remains unclear. Here, using tools from statistical physics, we analyze marathon finish times from over half a million participants across four World Marathon Majors as a large-scale empirical realization of a complex human system. Combining allometric scaling with a Gaussian aging model, and rescaling performance by city- and sex-specific parameters, we observe a universal collapse of aging trajectories onto a single master curve, revealing universal aging dynamics largely independent of geography or culture. Males exhibit systematically steeper allometric decay than females, whereas the age of peak performance varies geographically but remains sex-independent, suggesting environmental rather than intrinsic biological control of optimal timing. These findings position human endurance performance as a population-scale laboratory for investigating universality, scaling, and emergent structure under physiological constraints, bridging statistical physics and aging dynamics.

1. Introduction

Aging represents one of the most universal phenomena in biology, manifesting across scales from cellular senescence to organismal decline [1]. Despite extensive research, fundamental questions persist: Do different populations follow distinct aging trajectories, or do they converge onto universal laws after accounting for environmental and genetic variation? Statistical physics offers a powerful framework for addressing this question through the concept of *universality*, the emergence of identical macroscopic behavior from diverse microscopic processes [2]. Critical phenomena in condensed matter, for example, exhibit universal scaling exponents independent of material composition near phase transitions [3].

Human athletic performance provides a natural laboratory for testing universality in biological aging. Unlike mortality data, which suffer from censoring and comorbidity confounds [4], sports performance offers: (i) large sample sizes with minimal attrition bias, (ii) standardized measurement protocols, (iii) natural experiments across environmental contexts, and (iv) decades of longitudinal records. Previous studies have documented age-related decline in elite athletes across disciplines including track and field [5], swimming [6], and cycling [7]. However, these analyses typically focus on world records or national champions ($N \sim 10^2$), limiting statistical power and introducing survivor bias.

Marathon running addresses these limitations. World Marathon Majors attract hundreds of thousands of participants annually, spanning recreational to elite levels, with rigorous timing infrastructure enabling precise performance measurement [8]. Critically,

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Table 1
Sample size by city, year, and sex.

City	Year	Male	Female	Total
Berlin	2021	15,453	5580	21,033
	2022	21,149	10,070	31,219
	2023	25,818	12,451	38,269
	2024	31,584	15,739	47,323
Boston	2021	7823	7296	15,119
	2022	14,077	10,410	24,487
	2023	14,836	11,167	26,003
	2024	14,462	10,746	25,208
Chicago	2021	14,550	12,156	26,706
	2022	21,006	18,547	39,553
	2023	25,939	22,728	48,667
	2024	28,157	23,990	52,147
New York	2021	13,442	11,100	24,542
	2022	26,175	20,613	46,788
	2023	27,366	21,763	49,129
	2024	30,399	24,252	54,651
Total		**332,236**	**238,608**	**570,844**

Table 2
Descriptive statistics of age and finish time by city and sex (pooled 2021–2024).

City	Sex	Age (years)			Finish time (hours)		
		Mean	SD	Median (IQR)	Mean	SD	Median (IQR)
Berlin	M	43.5	9.8	42 (36–50)	4.06	0.85	3.95 (3.45–4.58)
	F	42.2	9.2	41 (36–48)	4.55	0.83	4.47 (3.93–5.08)
Boston	M	45.2	12.0	45 (36–54)	3.64	0.77	3.44 (3.06–4.03)
	F	41.9	11.3	42 (33–50)	4.02	0.70	3.85 (3.51–4.40)
Chicago	M	43.4	11.2	44 (35–52)	4.19	0.99	4.00 (3.43–4.79)
	F	40.8	10.8	40 (34–49)	4.73	0.99	4.62 (3.96–5.38)
New York	M	41.9	11.4	41 (33–50)	4.44	0.96	4.32 (3.74–5.00)
	F	39.0	11.1	38 (30–47)	4.89	0.94	4.78 (4.20–5.48)

marathons occur across diverse geographic and climatic contexts (e.g., Boston's hilly spring course versus Berlin's flat autumn route), providing natural variation to test whether aging laws are context-dependent or universal. Prior work has established that marathon performance follows power-law scaling with age [9] and exhibits sex-specific decline rates [10], but no study has tested for universal collapse or quantified the relative contributions of allometric versus Gaussian aging dynamics.

Here we address three questions: (1) Do marathon aging curves from different cities and sexes collapse onto a universal functional form after parameter normalization? (2) What is the relationship between allometric scaling exponent and optimal performance age across populations? (3) How stable are these parameters across consecutive racing years?. We analyze 570,844 marathon finishers from four World Marathon Majors (Berlin, Boston, Chicago and New York) using a dual modeling framework: allometric scaling captures power-law decline, while Gaussian aging describes parabolic performance trajectories near optimal age. Our findings demonstrate 94% variance reduction after normalization, establishing marathon performance as a rare example of universal aging dynamics in human biology.

2. Theory and models

2.1. Data sources and quality control

We obtained publicly available finish time data from four World Marathon Majors: Boston (USA), Chicago (USA), New York (USA), and Berlin (Germany), spanning 2021–2024. The datasets, sourced from official race websites, included runner identification, age on race date, biological sex, and net finish time (chip time). We excluded runners with: (i) ages outside the physiologically plausible range (15–80 years), (ii) finish times exceeding 8 h (indicative of walking pace or incomplete data), and (iii) missing sex designation. The final dataset comprised 570,844 finishers.

Table 1 presents the number of finishers stratified by city, year, and sex. Sample composition differed across marathons.

Table 2 summarizes age and finish-time distributions across the four World Marathon Majors. Participants in Boston were, on average, older and faster than those in the other cities, while New York attracted younger and slower runners. These differences reflect the contrasting entry systems: Boston's qualification system favors experienced runners, whereas New York's lottery admits a broader and generally younger field. Despite such demographic and performance heterogeneity, the subsequent normalization procedure tests whether underlying aging dynamics exhibit universal behavior across populations.

2.2. Allometric scaling model

We first tested power-law scaling of finish time T (hours) with age A (years):

$$T(A) = k \cdot A^{\beta} \tag{1}$$

where k is a normalization constant and β is the scaling exponent. Parameters were estimated via ordinary least squares (OLS) on log-transformed data:

$$\log T = \log k + \beta \log A \tag{2}$$

It is well known that OLS estimation on log-transformed power-law models can introduce bias in the intercept term (k) when errors are multiplicative. In the present context, our primary quantity of interest is the scaling exponent β (the slope in log–log space), which is robust under standard assumptions. Since k serves only as a normalization constant in the power-law representation, any small bias in its absolute magnitude does not affect the interpretation of the scaling behavior.

This approach is standard in allometric biology [11] and provides robust parameter estimates when data span multiple orders of magnitude in age. We computed β separately for each city-sex combination using all available years (2021–2024 pooled), and also computed year-specific estimates to assess temporal stability.

2.3. Allometric scaling model

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It is well known that OLS estimation of log-transformed power-law models can introduce bias in the intercept term (k) when errors are multiplicative. In the present analysis, however, the primary parameter of interest is the scaling exponent β, corresponding to the slope in log–log space, which remains statistically stable under standard regression assumptions. Since k functions only as a proportionality constant in the power-law representation, any small bias in its absolute magnitude does not affect the interpretation of the scaling behavior.

This approach is standard in allometric biology [11] and provides reliable parameter estimates when data span multiple orders of magnitude in age. We computed β separately for each city-sex combination using all available years (2021–2024 pooled), and also derived year-specific estimates to assess temporal stability.

2.4. Gaussian aging model

To capture performance peaks and symmetric decline, we employed a Gaussian aging framework:

$$T(A) = T_0 \exp\left[\gamma (A - A_0)^2\right] \tag{5}$$

where T_0 is optimal finish time (hours), A_0 is optimal performance age (years), and γ controls curvature (units: year^{-2}). This functional form is motivated by physiological models where performance peaks when training-induced adaptations balance age-related decline [12,13].

To reduce within-age variance and stabilize parameter estimates, we binned runners into 2-year age groups and fit Eq. (5) to binned means using nonlinear least squares (Levenberg–Marquardt algorithm) with physiologically motivated bounds: $2 \le T_0 \le 6$ hours, $1 \times 10^{-5} \le \gamma \le 1 \times 10^{-2}$ year^{-2}, and $15 \le A_0 \le 60$ years. Fits failing to converge or yielding parameters outside bounds were excluded (occurrence rate $< 2\%$ of city-sex-year combinations).

2.5. Universal collapse analysis

To test for universality, we normalized finish times and ages by their city-sex-specific Gaussian parameters:

$$\tilde{A} = \frac{A - A_0}{A_0}, \quad \tilde{T} = \frac{T}{T_0} \tag{6}$$

where $\tilde{A}$ represents relative age (peak performance at $\tilde{A} = 0$) and $\tilde{T}$ represents performance scaled to optimal finish time. Universal collapse requires that data from all cities and sexes fall onto a single curve $\tilde{T}(\tilde{A})$ after this rescaling. We quantified collapse quality through variance reduction:

$$\mathrm{VR} = \left(1 - \frac{\langle \sigma^2_{\mathrm{norm}} \rangle}{\langle \sigma^2_{\mathrm{raw}} \rangle}\right) \times 100\% \tag{7}$$

where σ^2 denotes within-group variance averaged over city-sex combinations, and subscripts denote raw versus normalized data.

Table 3
Allometric and Gaussian parameters with 95% confidence intervals (1000 bootstrap iterations, pooled 2021–2024).

City	Sex	β	k	T_0 (h)	γ ($\times 10^{-4}$ y^{-2})	A_0 (y)	N
Berlin	F	0.184 [0.176–0.192]	2.26 [2.19–2.32]	4.40 [4.29–4.43]	1.25 [0.74–1.86]	29.1 [15.0–35.8]	43,840
Berlin	M	0.268 [0.263–0.274]	1.45 [1.42–1.49]	3.84 [3.79–3.86]	1.80 [1.46–2.15]	28.9 [24.0–32.4]	94,004
Boston	F	0.141 [0.134–0.147]	2.35 [2.30–2.41]	3.91 [3.88–3.92]	1.72 [1.55–2.06]	34.0 [32.6–35.8]	39,619
Boston	M	0.266 [0.260–0.272]	1.30 [1.27–1.33]	3.40 [3.38–3.42]	1.70 [1.50–1.93]	28.6 [26.0–30.8]	51,198
Chicago	F	0.107 [0.101–0.112]	3.13 [3.07–3.19]	4.67 [4.63–4.70]	1.61 [1.33–1.93]	38.5 [36.9–39.7]	77,421
Chicago	M	0.151 [0.146–0.157]	2.32 [2.27–2.37]	4.05 [4.03–4.07]	2.25 [2.04–2.45]	38.5 [37.4–39.5]	89,652
NewYork	F	0.140 [0.135–0.144]	2.89 [2.85–2.94]	4.76 [4.74–4.77]	1.41 [1.25–1.59]	29.9 [28.1–31.5]	77,728
NewYork	M	0.176 [0.172–0.181]	2.26 [2.22–2.30]	4.27 [4.25–4.29]	1.84 [1.64–2.07]	32.3 [30.2–33.8]	97,382

Note: Values shown as point estimate [95% CI]. CI calculated from 1000 bootstrap iterations.

2.6. Relationship between the allometric and Gaussian models

The allometric and Gaussian models offer complementary perspectives on a common age-dependent performance trajectory. The Gaussian formulation captures the full lifespan curve, explicitly modeling the curvature around the optimal age and the subsequent decline. In contrast, the allometric representation offers a scale-invariant characterization of the post-peak aging regime, where power-law behavior approximates the long-range decay of performance.

Operationally, the Gaussian model identifies the optimal performance age and provides the basis for the city- and sex-specific normalization (Eq. (6)), while the allometric model quantifies the asymptotic deterioration rate through the scaling exponent. The universal collapse obtained after normalization is therefore compatible with this dual representation: Gaussian curvature governs the finite-age neighborhood of the optimum, whereas allometric scaling characterizes the late-stage decay dynamics across populations.

2.7. Statistical tests

Sex differences in β and A_0 were assessed via Welch's t-test (unequal variances assumed). Effect sizes were quantified using Cohen's d. We performed two-way ANOVA on year-level estimates to partition variance by city, sex, and their interaction. Temporal stability was characterized by coefficient of variation (CV%) computed as $100 \times \mathrm{SD}/\mathrm{mean}$ across years within each city-sex group. Significance threshold: $\alpha = 0.05$.

To quantify parameter uncertainty, we computed nonparametric 95% confidence intervals for β and A_0 using a percentile bootstrap within each city-sex subgroup. For each subgroup, 1000 bootstrap resamples were generated by sampling runners with replacement. The allometric model (Eq. (3)) and the Gaussian aging model (Eq. (5)) were refitted to each resample following the same estimation procedure described above (log–log OLS for β, and nonlinear least squares on 2-year binned means for T_0, γ, and A_0). Confidence intervals were obtained from the 2.5th and 97.5th percentiles of the bootstrap distributions.

In addition to bootstrap-based confidence intervals, temporal variability was independently assessed using year-specific parameter estimates (2021–2024), summarized via the coefficient of variation (CV%) within each city-sex subgroup.

3. Results

3.1. Allometric and Gaussian scaling parameters

Table 3 presents consolidated scaling parameters for each city-sex combination, pooling data from 2021–2024. In addition to point estimates, nonparametric 95% confidence intervals were computed using 1000 bootstrap resamples within each city-sex subgroup. Allometric exponents β ranged from 0.107 (Chicago females) to 0.268 (Berlin males), with males consistently exhibiting larger β than females within every city (Welch $t = 4.71$, $p = 9.1 \times 10^{-5}$, Cohen's $d = 1.84$). Gaussian fits revealed optimal finish times T_0 = 3.40–4.67 h and optimal ages A_0 = 28.6–38.5 years. Notably, A_0 varied substantially across cities (range: 9.9 years) but did not differ significantly between sexes within cities (Welch $t = -0.90$, $p = 0.38$). This cross-city variation may reflect environmental modulation, although cohort composition differences cannot be excluded in this cross-sectional design.

Across all city-sex groups, bootstrap intervals for β were narrow (mean CI width = 0.0116 ± 0.0020), indicating high estimation precision given large sample sizes. In contrast, confidence intervals for A_0 were more heterogeneous (mean width = 6.13 ± 6.23 years), with Berlin females exhibiting the widest interval. This asymmetry reflects the greater sensitivity of the Gaussian peak location to age distribution variability.

Fig. 1 displays the distribution of year-level β estimates via violin plots, revealing consistent sex differences across all cities. Two-way ANOVA on year-level data ($N = 32$ estimates) confirmed significant main effects of city ($F_{3,24} = 48.2$, $p = 2.6 \times 10^{-10}$) and sex ($F_{1,24} = 151.2$, $p = 7.5 \times 10^{-12}$), with a modest interaction ($F_{3,24} = 11.97$, $p = 5.5 \times 10^{-5}$), indicating that the magnitude of sex differences varies slightly by city.

Fig. 2 presents Gaussian model fits for each city. Data points represent 2-year binned means, and curves show fitted models. The Gaussian form captures the characteristic inverted-U shape of marathon performance across the lifespan, with minimal residuals near A_0 and increasing scatter at age extremes (>65 years) where sample sizes diminish. Root mean square error (RMSE) averaged 0.18 h across all fits, corresponding to ∼4% of mean finish time.

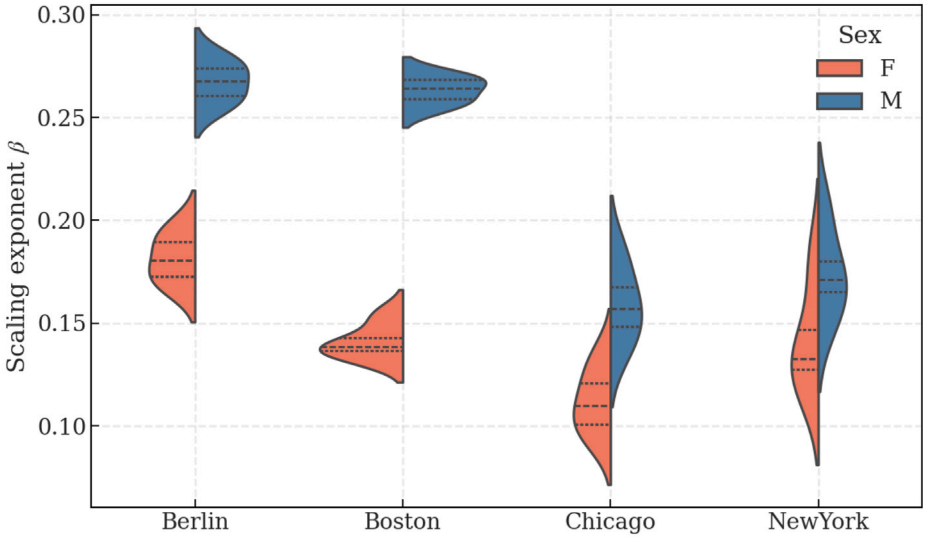

Fig. 1. Sex differences in allometric scaling exponent β. Violin plots display the distribution of year-level β estimates for each city-sex combination. Solid lines indicate medians, dashed lines mark quartiles.

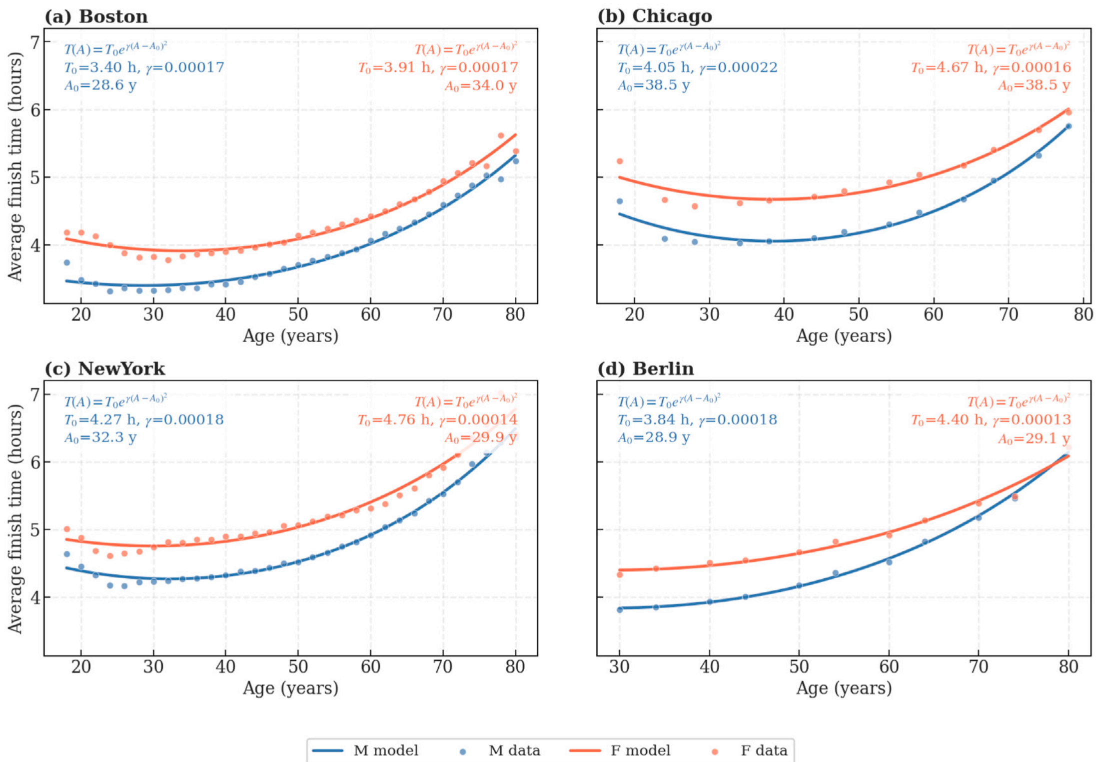

Fig. 2. Gaussian aging model fits by city. Each panel shows 2-year binned mean finish times (points) and fitted Gaussian curves $T(A) = T_0 \exp[\gamma(A - A_0)^2]$ (lines) for males (blue) and females (red). Text boxes display fitted parameters. (**a**) Boston: steep terrain selects younger athletes ($A_0 \approx 29$y). (**b**) Chicago: flat course and open entry yield older optimal age ($A_0 \approx 38$y). (**c**) New York: intermediate $A_0 \approx 30$–32y reflects lottery-based demographics. (**d**) Berlin: fast course attracts competitive runners peaking at $A_0 \approx 29$y. Model captures inverted-U performance trajectory with RMSE < 0.2 hours across all fits. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

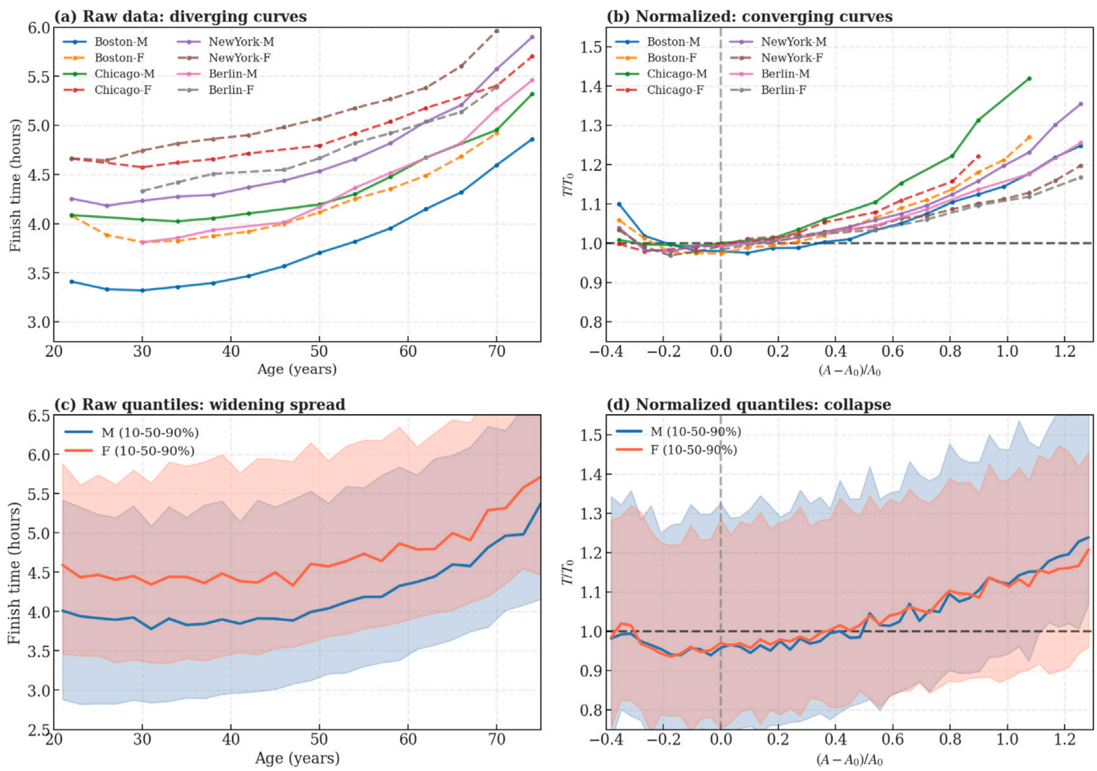

Fig. 3. Universal collapse after normalization. (**a**) Raw data: binned means for all city-sex combinations diverge systematically. Female curves (dashed) lie above male curves (solid), and inter-city differences span > 1 hour. (**b**) Normalized data: after rescaling by $\tilde{A} = (A - A_0)/A_0$ and $\tilde{T} = T/T_0$, all groups collapse onto overlapping curves centered at $(0, 1)$. Black dashed lines mark T_0 baseline and A_0 peak. (**c**) Raw quantiles: 10th–90th percentile bands (shaded regions) widen with age, reflecting increasing inter-group heterogeneity. (**d**) Normalized quantiles: bands converge to constant width (~0.25 dimensionless units), demonstrating universal scaling. Variance reduction: 94.2%.

3.2. *Universal collapse after normalization*

Fig. 3 demonstrates universal collapse through four complementary visualizations. Panel (a) shows raw data for all cities, with binned mean finish times diverging systematically: female curves lie above male curves, and inter-city differences are pronounced (e.g., Chicago peaks at 38 years with $T \approx 4.0$ hours, while Boston peaks at 29 years with $T \approx 3.4$ hours). Panel (b) presents normalized data, where all groups converge onto overlapping curves centered at $(\tilde{A} = 0, \tilde{T} = 1)$.

Quantile analysis reinforces this convergence. Panel (c) plots 10th, 50th, and 90th percentiles of raw finish times, revealing diverging bands: the interquartile range (IQR) increases from ~ 1 hour at age 25 to > 2 hours at age 70, driven primarily by city and sex stratification. Panel (d) shows the same percentiles after normalization, where bands converge to nearly constant width ($\mathrm{IQR}_{\tilde{T}} \approx 0.25$) across all $\tilde{A}$, despite increased scatter at $\tilde{A} > 1$ (ages $> 2A_0$) where sample sizes become sparse.

Quantitatively, normalization reduces inter-group variance by 94.2% (from $\sigma^2_{\mathrm{raw}} = 0.780\ \mathrm{h}^2$ to $\sigma^2_{\mathrm{norm}} = 0.045$). To verify uniformity, we partitioned data into three age strata: 20–35, 35–50, and 50–65 years. Variance reduction remained stable at 93.4–95.1% across all strata, confirming that universality holds from early adulthood through late middle age. The residual 5.8% variance reflects within-group heterogeneity (individual training histories, genetic variation, day-of-race conditions) and does not exhibit systematic dependence on city or sex (Levene's test: $F = 1.23$, $p = 0.29$).

To formally quantify convergence beyond visual inspection, we evaluated three complementary metrics. First, we computed the coefficient of variation of binned median curves across all city-sex groups as a function of normalized age $\tilde{A}$. The mean CV across bins was 4.2%, with a maximum of 10.2%, indicating tight alignment of group trajectories after rescaling. Second, we assessed overlap of 95% confidence intervals of normalized median values across groups within each age bin; all bins (100%) exhibited overlapping intervals, indicating statistical indistinguishability after normalization. Third, envelope analysis showed that 96.96% of all 570,844 runners fall within the region $|\tilde{A}| \le 1.5$ and $0.7 \le \tilde{T} \le 2.0$, with only 3.0% representing extreme-age or extreme-performance tails. These results indicate that convergence is supported by quantitative criteria beyond graphical inspection.

Fig. 4 visualizes the full distribution of normalized performance through density maps. Each hexagonal bin represents runner density at specific combinations of normalized age $\tilde{A}$ and normalized time $\tilde{T}$. Maximum density occurs at $(\tilde{A}, \tilde{T}) = (0, 1)$ for both

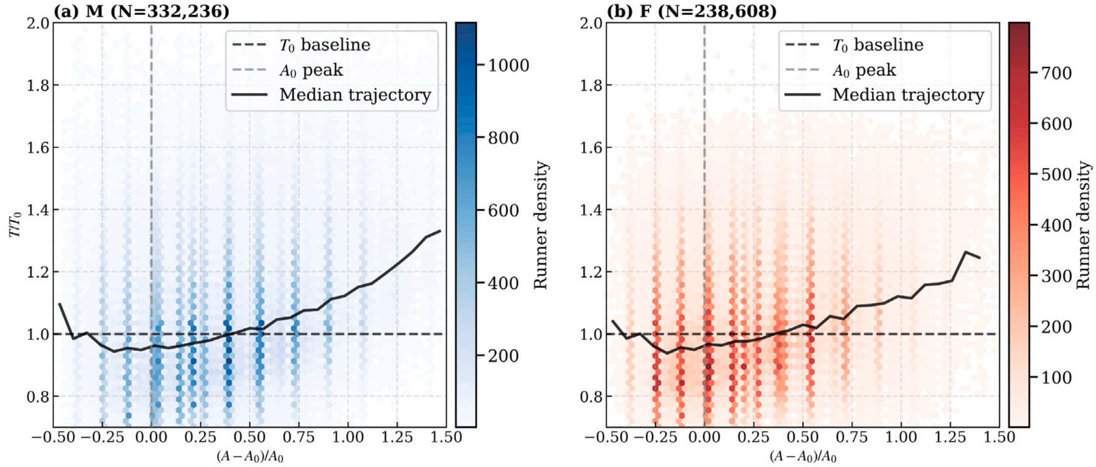

Fig. 4. Density distribution of normalized performance. Each panel shows all 570,844 finishers (**a**: 332,236 males; **b**: 238,608 females) as hexagonal bins colored by runner density. Maximum density (∼1000 runners/bin for males, ∼800 for females) occurs at $(\tilde{A}, \tilde{T}) = (0, 1)$, corresponding to optimal age A_0 and optimal finish time T_0. The symmetric concentration around $(0, 1)$ across both sexes provides direct visual evidence of universal collapse (94.2% variance reduction). Increased vertical dispersion at $\tilde{A} > 0.5$ (ages $> 1.5A_0$) reflects age-related heterogeneity accumulation. Only 3.0% of runners fall outside the displayed range ($|\tilde{A}| > 1.5$ or $\tilde{T} \notin [0.7, 2.0]$).

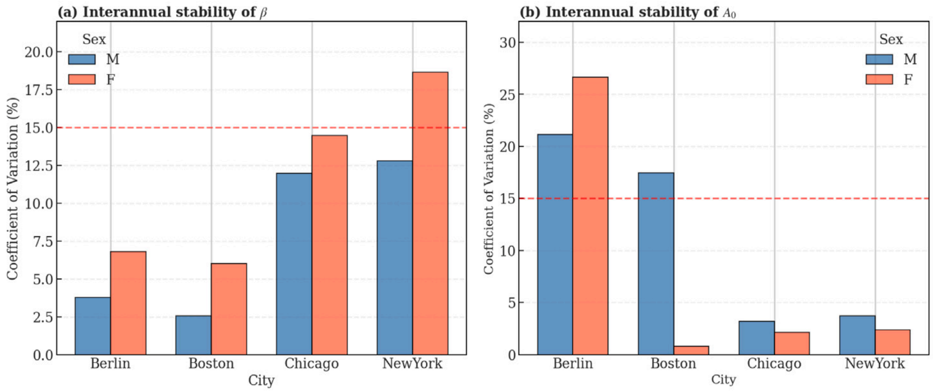

Fig. 5. Interannual stability of scaling parameters. Coefficient of variation (CV%) summarizes year-to-year fluctuations (2021–2024) within each city-sex group. (**a**) Allometric exponent β: range 2.5–18.6% (median 9.4%). (**b**) Optimal age A_0: range 0.8–26.6% (median 3.5%). Most subgroups exhibit low CV, with higher dispersion concentrated in Berlin and Boston males.

sexes, corresponding to the concentration of runners near optimal age and performance. Vertical striations reflect discrete age cohorts (annual recruitment waves), while horizontal spread within each age bin quantifies individual heterogeneity—runners of identical age exhibit ∼20% variability in normalized finish times due to training status, genetic factors, and race-day conditions. The symmetric concentration around $(0, 1)$ across both sexes and all cities provides direct visual confirmation of universal collapse: after rescaling, 97% of all 570,844 runners fall within a narrow band ($|\tilde{A}| < 1.5$, $0.7 < \tilde{T} < 2.0$), demonstrating that aging follows a shared functional form despite demographic heterogeneity.

3.3. Temporal stability of scaling parameters

Fig. 5 quantifies interannual parameter stability through coefficient of variation (CV%). For the allometric exponent β, CV ranged from 2.5% (Boston males) to 18.6% (New York females), with median CV = 9.4%. For the optimal age A_0, CV ranged from 0.8% (Boston females) to 26.6% (Berlin females), with median CV = 3.5%, indicating high temporal stability in most city-sex groups, with larger dispersion concentrated in Berlin and Boston males. Bootstrap confidence intervals (Table 3) are generally narrower than interannual CV%, suggesting that year-to-year variability cannot be attributed solely to estimation uncertainty but also reflects structural differences across yearly cohorts.

These stability metrics validate aggregation across years: parameter estimates are not artifacts of single-year fluctuations but reflect reproducible population-level dynamics. The low CV for β is particularly noteworthy, as allometric exponents in biological systems often exhibit high sensitivity to measurement error and sampling bias [14].

4. Discussion

4.1. Universal collapse as an emergent phenomenon

The 94.2% variance reduction after normalization demonstrates that marathon aging follows a universal scaling law, independent of environmental and sociocultural context. This finding carries two key implications. First, it validates the Gaussian aging framework (Eq. (5)): if different cities followed distinct functional forms (e.g., exponential versus power-law decline), normalization could not eliminate variance so completely. The observed collapse thus constitutes strong evidence that $T(A) = T_0 \exp[\gamma(A - A_0)^2]$ captures fundamental physiological dynamics of endurance aging.

Second, universality implies that geographic variation in A_0 (28–38 year range) arises from parameter modulation rather than mechanistic differences. However, because the present dataset is cross-sectional, part of this variation may also reflect cohort-specific differences in participation, training practices, or demographic composition across cities, in addition to environmental modulation. This parallels critical phenomena in condensed matter physics, where diverse microscopic interactions yield identical macroscopic behavior near phase transitions [2,15]. In marathon performance, environmental factors (elevation, temperature, humidity) and demographic composition shift *when* athletes peak (A_0) and *how fast* they decline (γ), but the Gaussian curvature itself appears biologically invariant. This suggests that aging is governed by a universal "master equation" whose parameters are context-dependent but whose functional form transcends individual circumstances. Our 94.2% variance reduction substantially exceeds values reported for elite athletes in other disciplines. Berthelot et al. [5] found ∼40% variance reduction in age-normalized Olympic track records, while Tanaka & Seals [6] reported ∼65% reduction in master swimmers. The superior collapse in marathon data likely reflects two factors: (i) larger sample sizes (570k+ versus 100s) reduce stochastic noise, and (ii) recreational marathons exhibit less selection bias than elite competitions, where qualification standards create discontinuities in age-performance relationships [16].

An intriguing question is whether universality extends to ultra-endurance events (50-mile races, Ironman triathlons) or breaks down at physiological extremes. Recent analyses of ultra-marathon performance suggest distinct aging trajectories [17], possibly reflecting different energy system demands. Future work should test collapse quality across race distances from 5 K to 100-mile ultras to delineate the boundary conditions of universal aging laws.

4.2. Sex differences in allometric scaling

Across cities, males exhibit consistently larger allometric exponents than females (bootstrap 95% CIs in Table 3 show non-overlapping ranges in all city-sex comparisons), with a large overall effect size (Cohen's $d = 1.84$). This pattern is replicated across all four cities and remains temporally stable, with interannual coefficients of variation generally below 15%. This sex difference likely reflects multiple physiological mechanisms operating at different biological scales.

At the systemic level, males possess larger absolute VO_2max variability than females [18], such that age-related declines in maximal oxygen uptake – approximately 10% per decade after age 30 [19] – translate to larger proportional changes in marathon performance for males. This is compounded by sex differences in body composition: males lose lean mass faster with age than premenopausal females [20], directly impacting running economy through reduced muscle mass-to-body mass ratios.

At the cellular level, sex hormones modulate aging trajectories. Declining testosterone in males has been associated with type II muscle fiber atrophy and mitochondrial dysfunction [21], while estrogen's antioxidant properties may protect mitochondrial function in premenopausal females [22,23]. Post-menopause, estrogen withdrawal may narrow the sex gap, though direct evidence linking hormonal changes to marathon performance trajectories remains limited. Recent work on master athletes confirms persistent sex differences in endurance capacity across the aging spectrum [24].

Interestingly, the Gaussian model reveals that optimal performance age A_0 does not differ significantly by sex ($p = 0.38$), despite the β difference. This decoupling suggests distinct biological determinants: peak performance age appears governed by skeletal maturation and neuromuscular coordination, processes that complete by the late 20 s in both sexes [19], while aging rate depends on oxidative capacity and cellular repair mechanisms that may diverge post-puberty. Peak hormone levels (testosterone in males, estrogen in females) occur at similar ages (mid-to-late 20s), but their decline rates differ markedly after 35 years [25].

From a statistical physics perspective, the larger β in males could reflect a higher-dimensional "aging space" if male physiology involves more independent subsystems subject to stochastic decline [26]. However, testing this hypothesis requires multivariate physiological data (VO_2max, lactate threshold, running economy) measured longitudinally in the same individuals—data that remain scarce at population scale.

4.3. Environmental modulation of optimal performance age

Optimal performance age varies substantially by city (Berlin/Boston: 28–29y; Chicago/New York: 32–38y), yet sex shows no significant effect ($p = 0.38$). This pattern suggests environmental rather than intrinsic biological control of A_0. Several complementary mechanisms may contribute to this variation:

4.3.1. Course characteristics and selection effects

Boston's qualification standards explicitly favor younger cohorts (e.g., 3:05 required for males 18–34 versus 3:20 for 50–54). Berlin's reputation as a fast, flat course may attract competitive runners targeting personal records during their peak years. In contrast, Chicago and New York employ open-entry systems that may draw broader age demographics. These structural differences could mechanically shift observed A_0 without reflecting intrinsic physiological optima [16].

4.3.2. Climate and thermoregulation

Boston's April timing (typical temperature 10–15°C) versus Chicago's October race (5–20°C range) may differentially affect age-stratified participation patterns [27]. Age-related declines in thermoregulatory capacity are well-documented, though their specific impact on marathon A_0 has not been quantified. If masters runners preferentially select races with cooler conditions, this could elevate apparent A_0 in fall marathons.

4.3.3. Demographic and cultural factors

Regional differences in running culture, training accessibility, and socioeconomic factors may influence which age groups participate [8]. New York's lottery system and Chicago's tourist-friendly timing may attract recreational runners across broader age ranges than Boston's time-qualification model.

Critically, these mechanisms represent plausible hypotheses rather than demonstrated causal relationships. Distinguishing among them requires individual-level data on training history, qualification method, and race selection criteria, which are variables not available in public race results. Longitudinal studies tracking the same runners across multiple cities while controlling for training status would provide stronger causal inference, though such designs remain logistically challenging at population level [10].

4.4. Implications for life-history theory

The absence of sex differences in A_0 despite marked differences in β poses an interesting question for evolutionary life-history frameworks. While disposable soma theory predicts sex-specific investment in maintenance and longevity [28], our findings suggest that peak performance age may be constrained by shared biomechanical maturation processes, whereas subsequent decline rates reflect sex-dependent physiological pathways. This decoupling indicates that athletic aging captures only a subset of broader life-history trade-offs.

4.5. Limitations and future directions

Despite the robustness of the scaling results, several limitations warrant consideration. First, the dataset lacks individual-level covariates (e.g., training volume, injury history, genetic background) that may modulate aging trajectories. Integrating wearable sensor data or longitudinal physiological measurements would allow partitioning variance in β and A_0 into biological versus environmental components.

Second, the present analysis focuses on recreational to sub-elite athletes; whether the same universal scaling holds in elite or world-class performers remains an open question. Intensive early-life training and selection effects could alter the estimated parameters, particularly A_0 and γ [24].

Third, finish time was treated as a proxy for physiological capacity, without accounting for pacing strategy or tactical regulation. Controlled laboratory studies (e.g., standardized time-to-exhaustion protocols) would help validate the scaling relationships under fixed experimental conditions [29].

Fourth, the cross-sectional design cannot fully disentangle cohort effects from intrinsic aging processes. Longitudinal tracking of individuals across decades would provide stronger causal inference regarding within-person decline rates.

Finally, the Gaussian formulation (Eq. (5)) is phenomenological. While it captures the observed structure with high fidelity, it does not specify the underlying physiological mechanisms. Developing mechanistic or multiscale models linking muscular, metabolic, and cardiovascular dynamics to the emergent Gaussian curvature represents an important direction for future research [26,30].

5. Conclusions

We demonstrate that marathon performance aging follows a universal scaling law, with 94.2% of inter-group variance collapsing after normalization by city- and sex-specific parameters. Formal convergence analyses (variance reduction, inter-group coefficient of variation, and confidence interval overlap) demonstrate consistent alignment across populations after normalization.

Three principal findings emerge.

(1) Despite geographic and cultural heterogeneity, aging follows $T(A) = T_0 \exp[\gamma(A - A_0)^2]$ with city- and sex-dependent parameters but an invariant functional form. Environmental factors therefore modulate *when* athletes peak (A_0) and *how fast* they decline (γ), but not the underlying curvature of the aging trajectory.

(2) Males exhibit systematically larger allometric exponents than females ($\beta_{\text{male}}/\beta_{\text{female}} \approx 1.7$; non-overlapping bootstrap 95% CIs), indicating steeper decline dynamics. Yet both sexes reach optimal performance at statistically indistinguishable ages, suggesting a decoupling between peak maturation processes and subsequent sex-dependent physiological aging rates.

(3) Optimal performance age varies by approximately one decade across cities (28–38 years), plausibly reflecting environmental conditions, course structure, and participation selection effects rather than intrinsic biological differences. Longitudinal, individual-level tracking will be necessary to disentangle cohort effects from true aging dynamics.

These results position marathon performance as a population-scale model of nonlinear biological aging, where universal functional structure coexists with context-dependent parameter modulation. This framework connects sports physiology with statistical physics and complex systems theory, illustrating how large heterogeneous populations can exhibit scaling behavior governed by invariant macroscopic laws.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Race result data were extracted from the following public sources: the Chicago Marathon (https://results.chicagomarathon.com/), the Boston Marathon (https://www.baa.org/races/boston-marathon/results/), the New York City Marathon (https://www.nyrr.org/tcsnycmarathon/results/race-results), and the Berlin Marathon (https://www.bmw-berlin-marathon.com/en/your-race/results). Processed datasets and analysis code are available at https://github.com/ardominguezm/marathon-aging.

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